

A Packing of 6 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 6 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 3.199811. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.110922, obtained by deleting 1 of the 7 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 6 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.199811 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 6$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 3.110922, obtained by deleting 1 of the 7 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 6/5 = 3.200000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N6.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
106660397313546085908731219212486633050074350550364248577747236304602700
156241164807790975513180029258618051404372393979110759461583331241367671
9682443829176991571084751763620132522642674340819
```

denominator:

```
333333333333853078529599013330275272736607594396941718649817623559956083
849686080862564193983506212558186877591282309880403058475248877549961894
466191558091399719571332352394945382547657350326
```

B Coordinates

6 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N6.txt`.

```
-220202867389 399768530284 -215402508838 -149887935975300943033 340600753365 296710635176 21106663394294549525043 535862409985 68143456741 261203141933
-16908546883 -435740555521 -548882522866 31766766032802153377 352857924073 -12427621722 -496897233295-53123315724 118733961950 -518053894691 -410838469137
337053168445 202267788603 -434006081845 -122273774550533405866 67764260765 613635894300 144983907744 -253342609728 -271366799405 150688637649 -60841721248
632653756415 483568455437 -81621832759 2227536701940808226927 97841354700 -17141019689 -299935727950 77021704693 -199483443935 482458652145 111577761103
54934009225 -83988001951 -422568381869 -32981035886627223056341 152938804157 -154854165467 38692326837022796853369 152588124857 -279605737168 305692433535
105706890603 27903850180 -8772392565 431885362622 64823611603 -56476352499 200446819406 421043589946 318414734477 -273400623728 -274704586017 -144770905371
-277191341349 142852807833 124420122757 15762774871469682726891 -278337626189 485735676493 -347497819507988656556 354357906025 361916844348 -58611293012
189577751280 -124379313229 10855247893 323208053178172836273285 -320669910245 -238862696550 31064229617086852721143 178811415488 -328622233393 603633333218
125112475121 -445539342114 -162340843930 15598930557361805253489 -639524427626 101356664698 -13287471680551914301 145925733269 39117698689 73435724995
523304915352 -330897466613 347140276415 -4222726895521630871640 247726311841 96748270650 -2664728045266693855984 202123211299 19268053743 -256189839032
143410168440 21613932919 220884645565 376776398329 203725140036 -252601419951 107448041399 515447903 -249570389838 441880023296 -51442945078 -426130842234
18305368247 -152935387005 259646277218 255098776414667292621563 -65339944387 -382927891014 -1869639444297242953588 268960940815 272648186808 -109566698722
-318647671831 -113728422738 274601294602 -871895139550935346257 197670152650 105869789268 -3173551786256248802080 168276886931 89388928679 -84316945005
254040930746 78428438345 -158790417652 -83386810234-350569551022 -46626373890 -189194426608 -42115775025389678665 -9156882845 -332321417909 170164653714
-244146572360 -257601869577 -393911750478 155113941120246494507 -161619586831 -264752974315 -18199795274896298478 461862635924 -15943280294 -331395519443
209243925360 87960606511 7645381071 -413090154710 -314860758612 -2001673170 -55025506411 447285789044-159155103414 -106033073777 -508892604985 312556094349
198766582331 -481516372595 202354058941 1338087692236747697255 179764530586 -294688129657 5377635211584305492674 46443708235 -399711360449 -166424554523
-46502161748 -2801859059 212492726391 386791009578 422368284488 700180232451 167645392623 -77035343041-230708889615 103882829708 -456848517082 363185517724
390483975674 282564351887 325317364670 4445713164342611742802 191328503707 -209139021347 -45672646479-49169532835 -70859059413 -321143191390 -388857491339
91769375241 289565168824 -448176706661 -2940213014667047619474 116046974438 187664464141 -130605676814313251857473 -148267746559 -269140366380 -613641168868
-218092865404 182106421345 154512438996 -4967499533477689709 -388020313156 -17786299997 243153796627 -48536496676 814235542286 -127015280455 31234194082
104642629080 68433115842 -555759147980 556322226957-206291558086 156277643006 -37050814395 235585267145420611090334 -68433329609 43485726540 -153669462778
209897073645 -688047205931 -127109184809 -9964384866351005940221 125484030928 663578384612 23789012703097236406288 189074605523 187666209101 25244088335
-652223105672 2787353670 -49615238894 131322667584 -204900195762 411409463120 -493909477817 5692298345283075150464 -60614329444 -158286138651 -209348305440
```